

End Semester Examination-November 2015

B. Tech. Course in Electronics & Telecommunication Engineering

Subject: Digital System Design (OLD Course)

Semester: VI

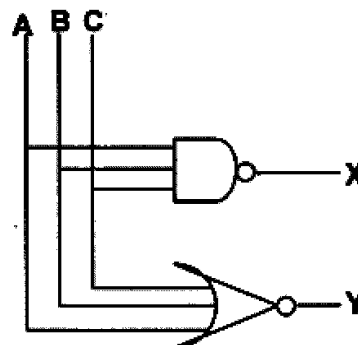
Duration: 03 Hrs

Max. Marks: 50

Instructions to the student:

1. Attempt any five questions.
2. Illustrate your answer with neat sketches, diagrams etc. wherever necessary.
3. Necessary data is given in the respective questions. If such data is not given, it means that the knowledge of that component is a part of examination.
4. If some part or parameter is noticed to be missing, you may appropriately assume and state it clearly in the answer-book.
5. All symbols and notations are having their usual meaning.

- Q.1. A] Using the timing diagram show the signals at output X and Y of the Figure below as function of the three inputs A, B, and C. Use all eight possible combinations of A, B, and C. 05



- B] Design a combinational circuit using a ROM. The circuit accepts a three-bit number and outputs a binary number equal to the square of the input number. 05

- Q.2. A] Using Boolean algebraic theorems, prove that 05

$$A + \bar{A} \cdot B + A \cdot \bar{B} = A + B$$

- B] Write a VHDL code for a 3: 8 line decoder. 05

(P.T.O.)

- Q.3. A] Using K-map, minimise the logic function in POS form and draw the logic diagram. 05
- $$f(A, B, C, D) = \Pi M(4, 6, 10, 12, 15)$$
- B] Write a VHDL code for 2:1 Multiplexer 05
- Q.4. A] Construct a 32:1 Multiplexer using two 16:1 multiplexer and one OR-gate. 05
- B] Write a VHDL code for J-K flip flop. 05
- Q.5. A] Design a 4-bit GRAY to BCD converter. 05
- B] With suitable examples, explain the concurrent and sequential statements used in VHDL programs. 05
- Q.6. A] What is a shift register? Classify different types of shift registers with the help of their truth table and timing diagrams. 05
- B] Write a VHDL code for EX-NOR gate 05
- Q.7. A] Design a 4-bit synchronous down counter using J-K flip flop. Draw the timing diagram for the counting process. 05
- B] Write a VHDL code for 4-bit UP-counter. 05
- Q.8. What is the difference between a PAL and PLA. Design a BCD to Excess-3 code converter using PLA. 10

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